

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I G. Manobhi ram(192311322) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Manobhi Ram

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

ANSWERS

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

A company experiencing heavy traffic during seasonal sales can use cloud auto-scaling and load balancing to maintain application performance.

Auto-scaling automatically increases or decreases the number of servers based on CPU usage, memory, or incoming requests. During high traffic, new virtual machines or containers are launched automatically, and when traffic decreases, extra resources are removed. This helps reduce infrastructure costs.

Load balancing distributes incoming user requests across multiple servers. Instead of sending all requests to one server, it ensures that each server handles an equal amount of traffic. This prevents server overload, reduces response time, and improves application availability.

Monitoring tools such as AWS CloudWatch, Azure Monitor, or Google Cloud Monitoring continuously track system performance, CPU utilization, memory usage, network traffic, and application health. If resource usage exceeds a predefined threshold, they trigger auto-scaling and alert administrators.

Real-world Example

During the Amazon Great Indian Festival or Flipkart Big Billion Days, millions of users access the websites simultaneously. Auto-scaling automatically launches additional cloud servers, while load balancers distribute customer requests evenly. As a result, customers experience fast page loading, fewer errors, and uninterrupted shopping.

Benefits

- Handles sudden traffic spikes automatically.
- Reduces downtime.
- Improves website speed and user experience.
- Optimizes cloud costs by using resources only when needed.
- Ensures high availability.

2)An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Sol) Organizations generate large amounts of backup data that must be stored safely and be available when needed. **Cloud storage** provides a secure, reliable, and cost-effective solution for storing backups.

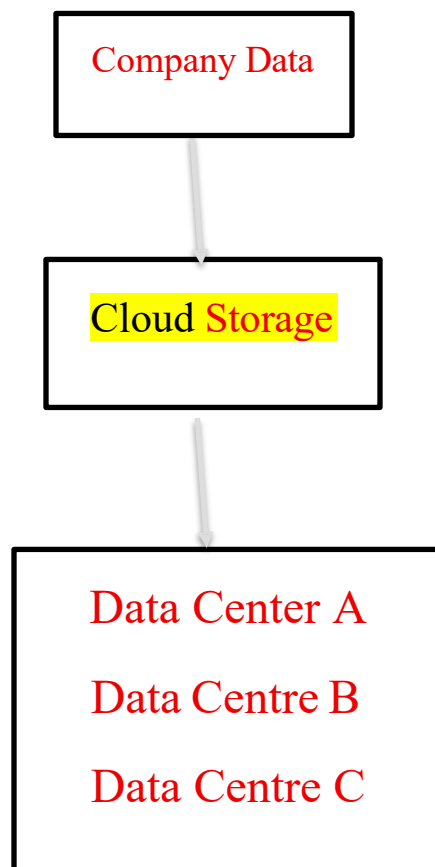
Cloud Storage Durability and Data Redundancy:

- Encryption at rest and in transit.
- Identity and Access Management (IAM).
- Multi-Factor Authentication (MFA).
- Role-Based Access Control (RBAC).

Audit logs for monitoring data accessEncryption and Access Control:

- **Encryption** protects data by converting it into an unreadable format during transmission and while stored in the cloud. Only authorized users with the correct encryption key can access the data.
- **Access control** uses user authentication, passwords, role-based access control (RBAC), and multi-factor authentication (MFA) to ensure that only authorized users can view or modify backup data.

Cloud Backup Diagram:



Cost Advantages over Traditional Storage:

- No need to purchase expensive storage hardware.
- Lower maintenance costs.
- Pay only for storage actually used.
- Automatic backup and scaling.
- Reduced disaster recovery expenses. **Disaster Recovery Use Case:** A company stores daily backups of its customer database in the cloud. If its office servers are damaged by a flood, fire, or cyberattack such as ransomware, the organization can quickly restore the latest backup from the cloud and resume business operations with minimal downtime and data loss

Cost Comparison Table:

Traditional Storage	Cloud Storage
Buy Hardware	Pay as You Use
High Maintenance	Low Maintenance
Limited Storage	Easily Scalable
Manual Backup	Automatic Backup

Security Technologies:

- AES-256 Encryption
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- SSL/TLS Secure Transmission

Conclusion:

Cloud storage offers high durability through data redundancy, strong security through encryption and access control, lower operational costs, and reliable disaster recovery, making it an ideal solution for modern backup and recovery systems.

3) A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Sol) A mobile application deployed globally must provide fast and reliable access to users in different countries. **Content Delivery Networks (CDNs), edge servers, and cloud infrastructure** help improve performance, reduce latency, and ensure high availability.

1. How CDNs Improve Performance and Reduce Latency

- A CDN stores copies of static content such as images, videos, JavaScript files, and CSS files on edge servers located around the world. Instead of downloading data from the original server, users receive content from the nearest edge server.
- This significantly reduces network latency and improves loading speed.

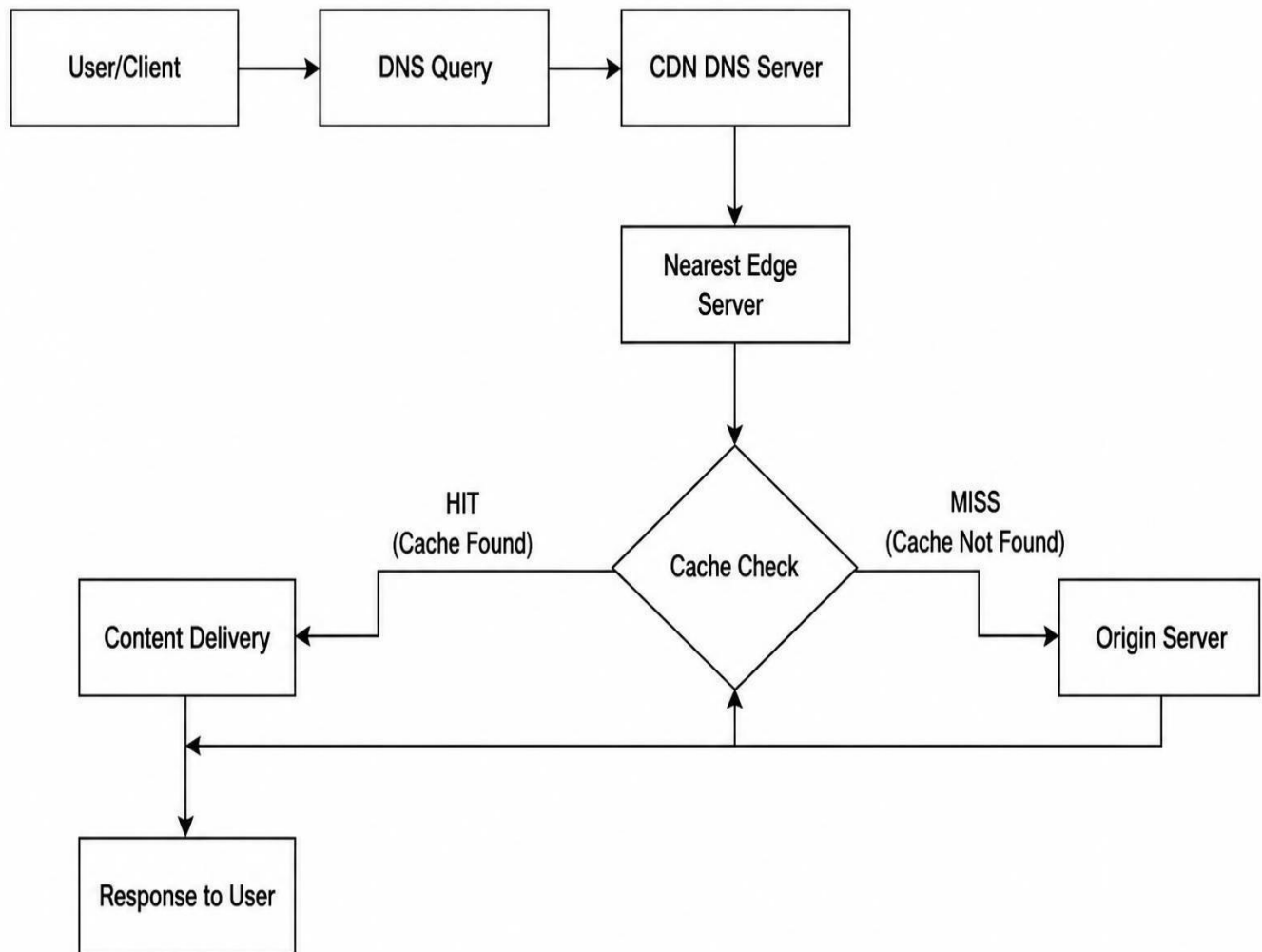
Edge Servers

- Edge servers are geographically distributed servers located close to end users. They reduce the physical distance data must travel, resulting in:
 - Faster response time.
 - Lower latency.
 - Reduced network congestion.
 - Better user experience.
- Cloud providers also deploy applications in multiple regions. If one region becomes unavailable, traffic is redirected to another healthy region, ensuring continuous availability.

Advantages:

- Faster response time.
- Lower latency.
- Reduced buffering.
- Smooth user experience.
- Lower workload on the central server.

CDN Architecture:



2. Cloud Providers Support Global Availability:

Cloud providers maintain multiple **data centers** across different countries.

They provide:

- Global data centers.
- Automatic traffic routing.
- Load balancing.
- Auto-scaling.
- Disaster recovery.
- High availability (24×7 uptime).

If one region becomes unavailable, traffic is automatically redirected to another healthy region, ensuring uninterrupted service.

3. Real-Time Streaming / Gaming Example

Consider **Netflix** or an online multiplayer game like **PUBG Mobile**.

- A user in **India** receives video/game data from the nearest CDN edge server in India.
- A user in **Europe** connects to a nearby European edge server.
- The cloud automatically balances traffic across multiple regions.
- During peak hours, additional servers are launched automatically.
- This reduces buffering, lag, and game latency while providing a smooth viewing or gaming experience.

Conclusion

CDNs, edge servers, and cloud services work together to reduce latency, improve performance, ensure global availability, and provide a seamless experience for users worldwide.

4)A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Sol) Financial institutions handle highly sensitive customer and transaction data. Therefore, they must follow strict **regulatory compliance, security standards, and data protection policies** when using cloud services.

1. Cloud Providers Support Regulatory Compliance:

Cloud providers offer built-in security controls and certifications that help organizations comply with government and industry regulations.

They support compliance by:

- Following international security standards (ISO 27001, PCI-DSS, SOC 2).
- Providing regular security audits.
- Maintaining compliance reports and documentation.

- Enforcing identity and access management (IAM).
- Offering backup and disaster recovery services.

This helps financial institutions meet legal and regulatory requirements.

2. Role of Logging, Monitoring, and Encryption:

Logging

- Records every user activity and system event.
- Tracks login attempts, transactions, and file access.
- Helps during audits and investigations.

Monitoring

- Continuously monitors cloud resources.
- Detects unusual activities, cyberattacks, and unauthorized access.
- Sends alerts to administrators for quick response.

Encryption

- Encrypts data both **at rest** and **in transit**.
- Protects customer account details and financial records.
- Only authorized users with encryption keys can access the data.

3. Ensuring Data Sovereignty:

Data sovereignty means data must remain within the country's legal boundaries.

A company can ensure this by:

- Selecting cloud data centers located within the required country.
- Configuring data residency policies.
- Restricting cross-border data transfers.
- Following local government regulations.
- Regularly auditing data storage locations.

This ensures compliance with national data protection laws

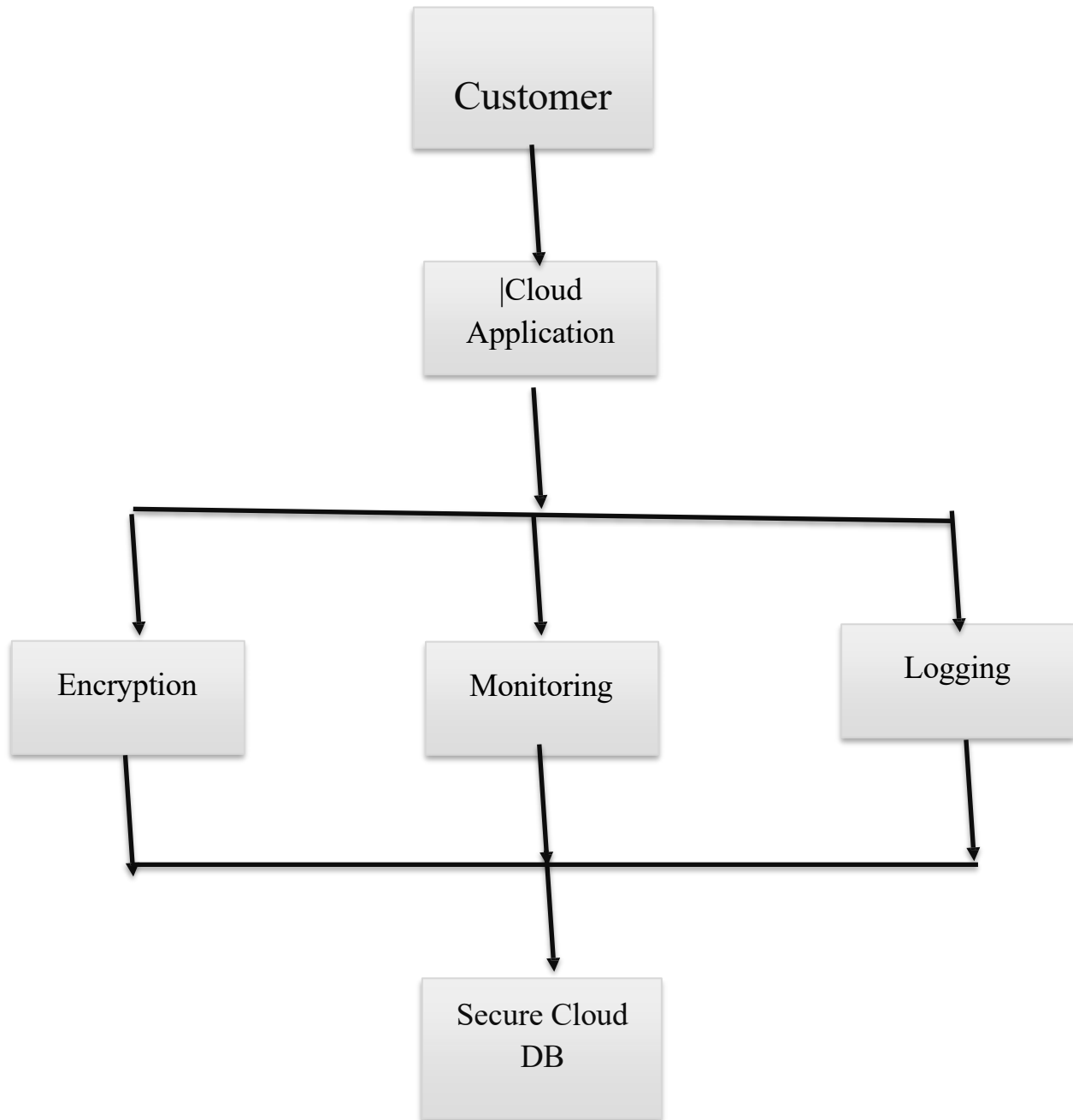
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4. Example – Sensitive Financial Transactions:

Consider an online banking system where customers transfer money between accounts.

- Every transaction is encrypted.
- All login attempts and money transfers are logged.
- Monitoring tools detect suspicious activities such as unusual login locations or large transactions.
- Customer financial data is stored in cloud data centers located within the country to satisfy data sovereignty requirements.
- If suspicious activity is detected, administrators receive immediate alerts and can take corrective action.

Security Flowchart:



Conclusion:

Cloud providers help financial institutions achieve **regulatory compliance**, **secure data through encryption**, **track activities using logging**, **detect threats with monitoring**, and **meet data sovereignty requirements**, ensuring secure and reliable financial services.

5)A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Sol)A development team can use **containerization** to package an application along with its dependencies into lightweight containers. Containers provide faster deployment, efficient resource utilization, and consistent performance across different environments.

1. Containers vs Virtual Machines

Containers	Virtual Machines
Share the host operating system	Each VM has its own operating system
Lightweight	Heavyweight
Fast startup (seconds)	Slow startup (minutes)
Low resource usage	High CPU and memory usage
Higher performance	Comparatively slower
Suitable for microservices	Suitable for running multiple OSs

Difference:

- Containers are lightweight because they share the host OS kernel.
- Virtual Machines require a complete guest operating system, consuming more CPU, RAM, and storage.
- Containers start much faster and allow more applications to run on the same hardware.

2. Kubernetes (Container Orchestration)

Kubernetes automates the deployment, management, and scaling of containers.

It simplifies deployment by:

- Automatically deploying containers.
- Scaling applications up or down based on demand.

- Restarting failed containers automatically.
- Load balancing traffic among containers.
- Performing rolling updates with minimal downtime.
- Managing container health and availability.

This reduces manual effort and improves application reliability.

3. Portability Across Cloud Environments

Containers package the application and all required libraries together.

Therefore, the same container can run on:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- Private Cloud
- Local Computer

This eliminates compatibility issues and supports "**Build Once, Run Anywhere.**"

4. Example – Microservices-Based Application

Consider an **online food delivery application (Swiggy/Zomato)**.

Each service runs in a separate container:

- User Service
- Login Service
- Restaurant Service
- Order Service
- Payment Service
- Notification Service
- Delivery Tracking Service

If the **Order Service** receives heavy traffic, Kubernetes automatically creates additional containers only for that service without affecting the others. This improves scalability, availability, and performance.

Conclusion

Containers provide lightweight, fast, and portable application deployment. Kubernetes automates deployment, scaling, and recovery, while microservices improve flexibility and maintainability, making containerization ideal for modern cloud applications.